

3D Empirical Model of the Midlatitude Ionospheric Trough Based on Swarm Observations

Máté Tomasik^{1,2,3}, Balázs Heilig^{1,2}

Email: tomasik.mate@epss.hu

¹HUN-REN Institute of Earth Physics and Space Science, Sopron, Hungary

²Karman Space Research Center, Obuda University, Budapest, Hungary

³ELTE Doctoral School of Earth Sciences, Budapest, Hungary

Introduction

The midlatitude ionospheric trough (MIT) is a region of significantly reduced electron density located equatorward of the auroral zone. Its position and structure vary with geomagnetic activity, local time, and solar conditions. In this work, we develop a model to predict both the location and the electron density profile of the MIT using regression techniques.

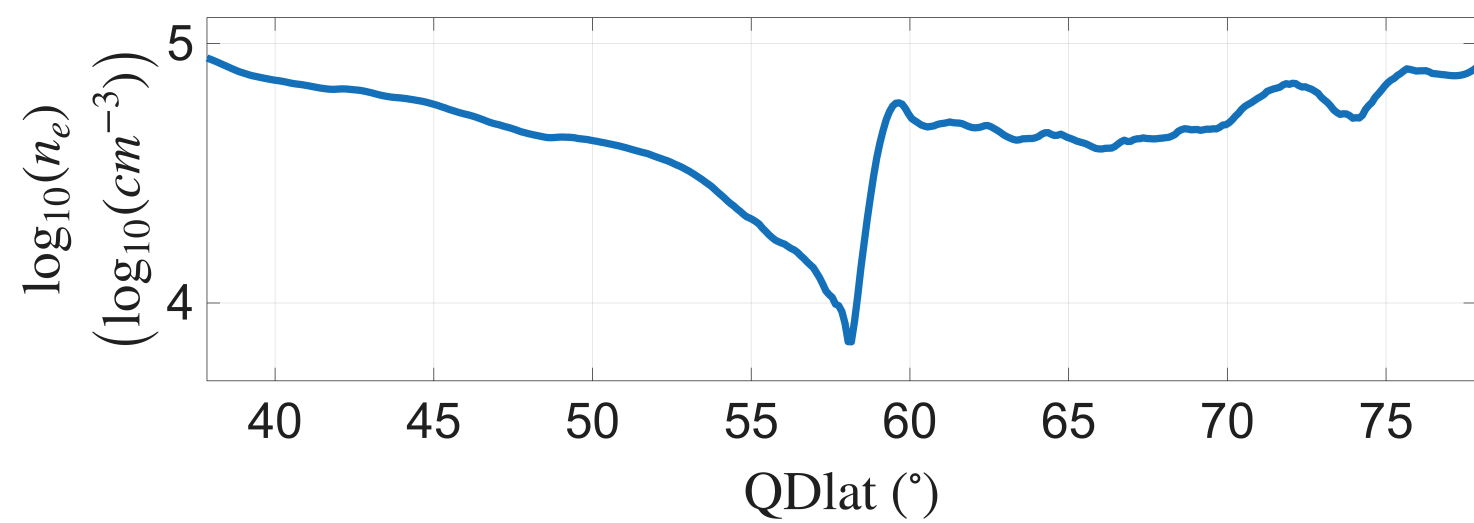


Fig. 1. Electron density profile of a midlatitude ionospheric trough (MIT) observed by Swarm-A satellite

Magnetosphere–Ionosphere Coupling

Electric potentials map along equipotential magnetic field lines, linking plasma structures in the ionosphere and magnetosphere.

The midlatitude ionosphere is strongly controlled by the interplay between **corotation** and **convection electric fields** and the **ExB drifts**, including subauroral polarization streams (SAPS) and subauroral ion drifts (SAID).

The trough is closely related to the **plasmopause** and the **plasmaspheric plume** in certain magnetic local times (MLT).

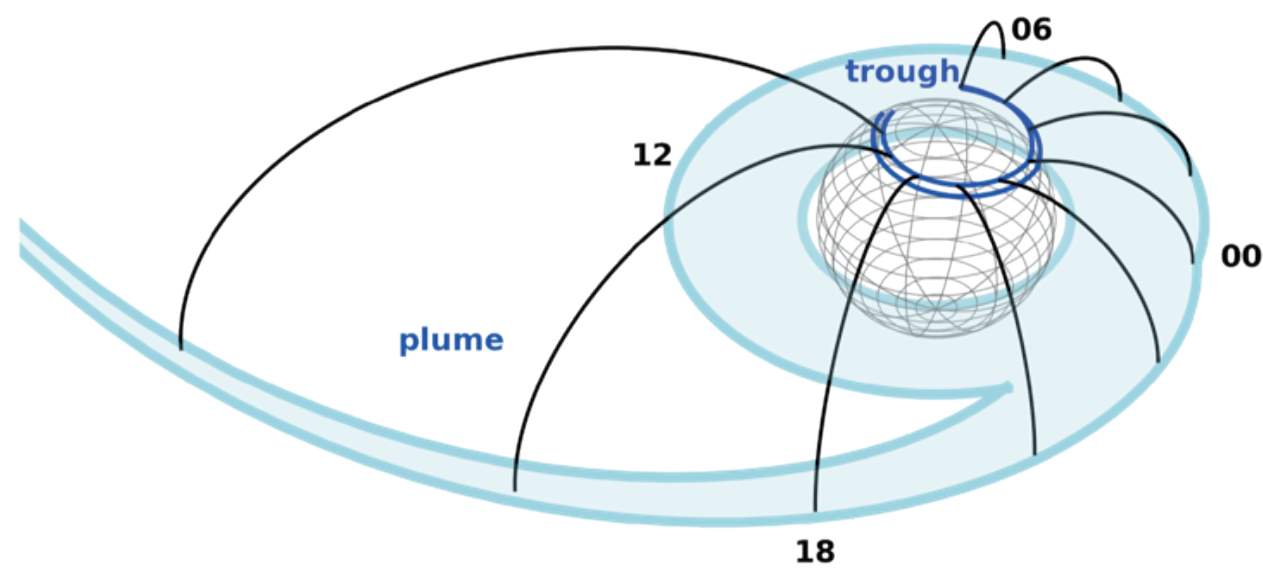
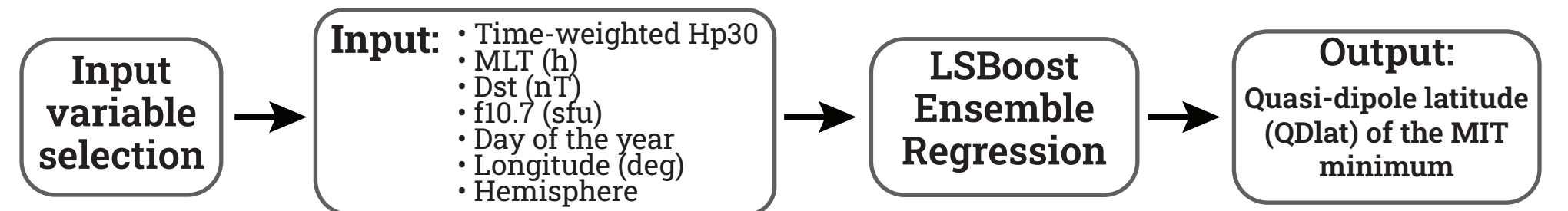


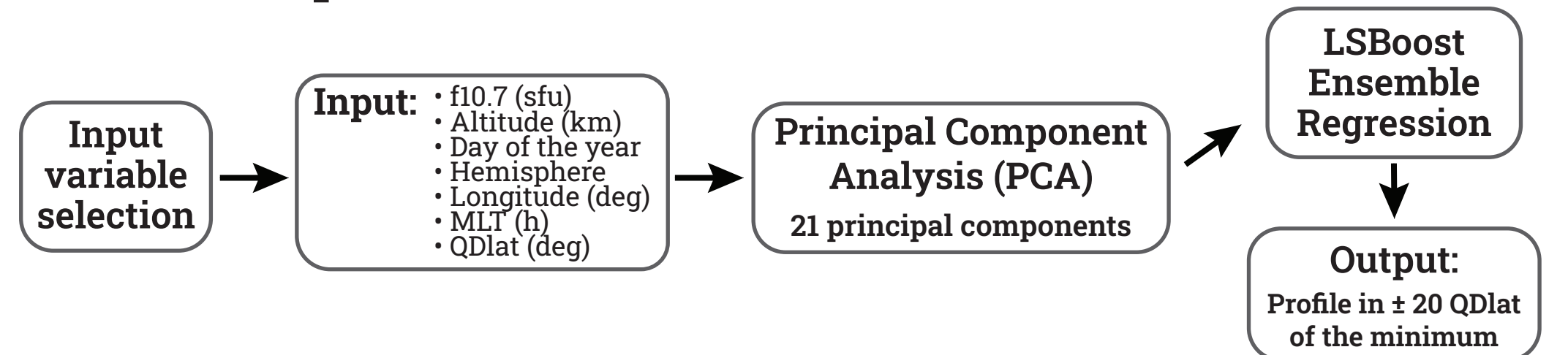
Fig. 2. Visualization of how the magnetic field lines link the trough to the plasmopause and to the plasmaspheric plume in certain magnetic local times

MOSAIC 3D MIT model

1: Location of the MIT minimum



2: The MIT profile



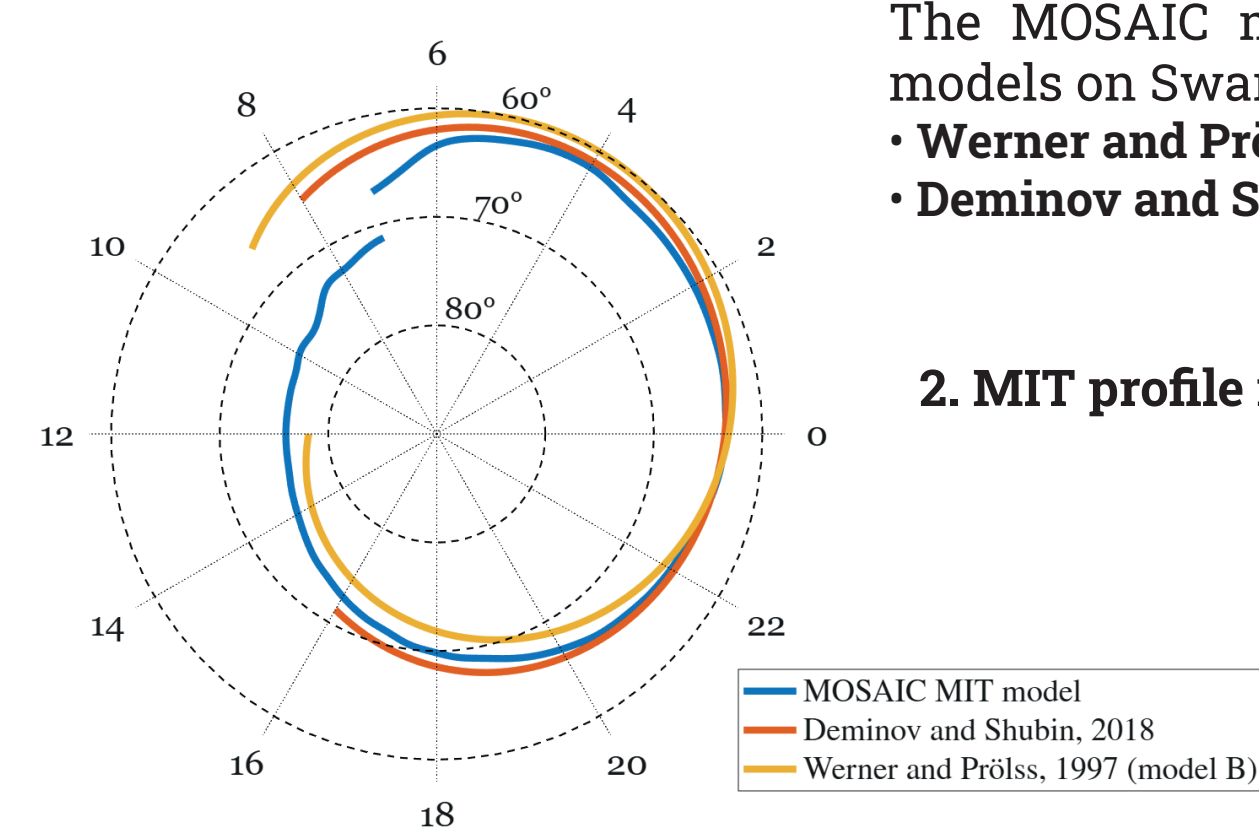
Validation of the model

Validation dataset: March and September of 2015 and 2020 (different phases of Solar Cycle 24).

1. MIT Minimum model: RMSE = 1.49°, MAE = 1.1°, $r^2 = 0.87$

The MOSAIC model outperforms the previous models on Swarm data:

- Werner and Prölss (1997) Model-B : RMSE = 3.53
- Deminov and Shubin (2018): RMSE = 2.01



2. MIT profile model: RMSE = 0.24 ($\log_{10}(n_e)$), MAE = 0.19 ($\log_{10}(n_e)$), $r^2 = 0.66$

Fig. 3. MIT minimum predictions of different models as a function of MLT

2024 May Storm and the Plasmopause

The model output is shown for the 2024 geomagnetic storms in L-shell – MLT space, projected onto the equatorial plane. The color scale represents $\log_{10}$ electron density, while the black line marks the predicted position of the MIT minimum.

The results are compared with the ELTE PLASMA empirical neural-network plasmopause model, with the modeled plasmopause shown in red. The MIT minimum and the plasmopause are generally co-located on the nightside, particularly between dusk and dawn.

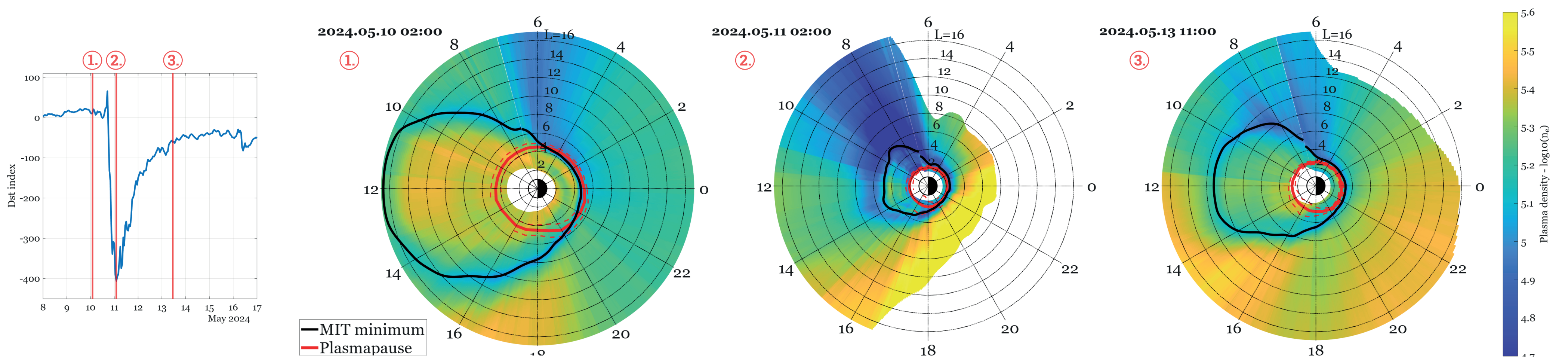


Fig. 3. Model output for the 2024 geomagnetic storms in L-shell – MLT space, projected onto the equatorial plane. First plot shows the Dst index values during the storm, the next three plots show the output for the selected times. Colors show $\log_{10}$ electron density, the black line marks the predicted MIT minimum, and the red line indicates the plasmopause from the ELTE PLASMA model.

The dataset

Electron density: ESA's Swarm L1 EFI measurements | MIT minimum locations: ESA L2 PRISM MIT product (extended).
The Hp30 index: Helmholtz Centre for Geosciences (GFZ) | Other space weather parameters: OMNIWeb.